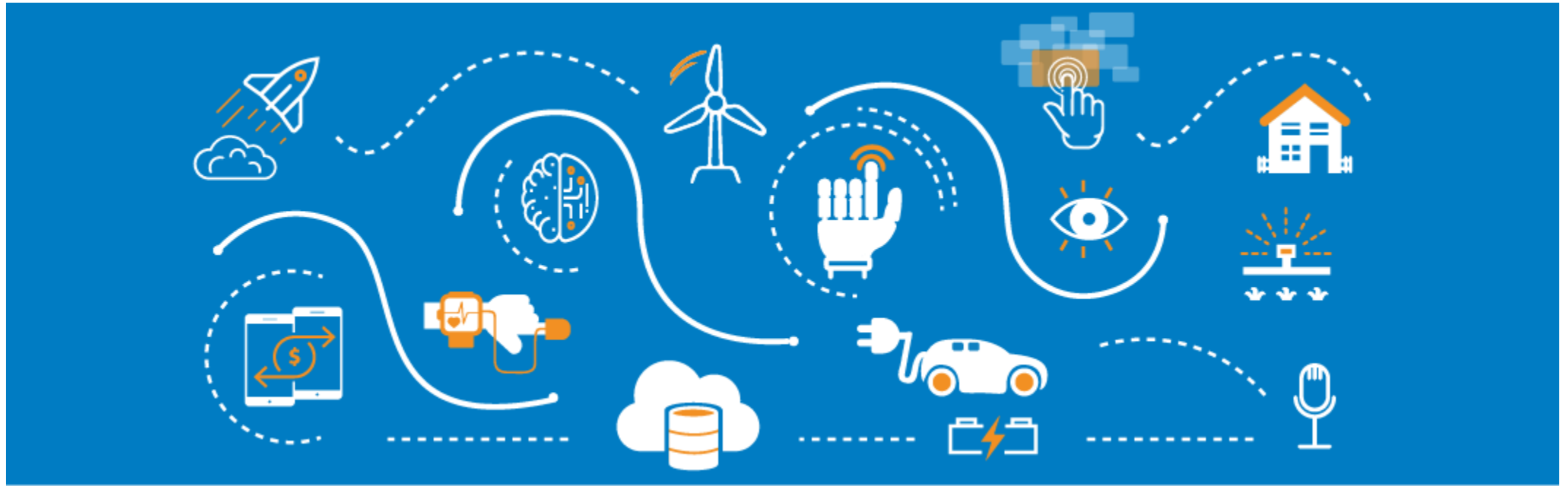




Computer Vision Fundamentals Part 1

Kien Tran

Contents

1 Introduction to Computer Vision

2 Numpy and Image Basics

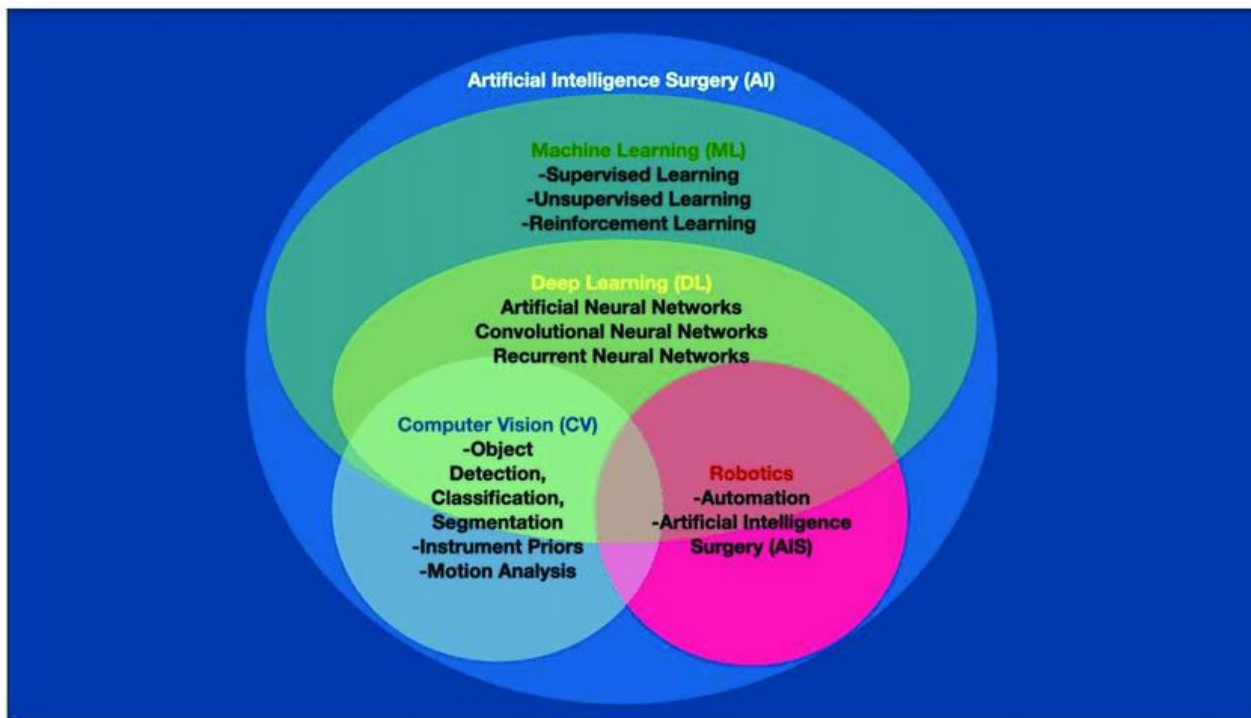
3 Image Basics With OpenCV

4 Image Processing

1. Introduction to Computer Vision

1.1. What is computer vision?

Enables computers to analyze images or video data, unlocking a multitude of applications across industries, from autonomous vehicles to facial recognition systems



1. Introduction to Computer Vision

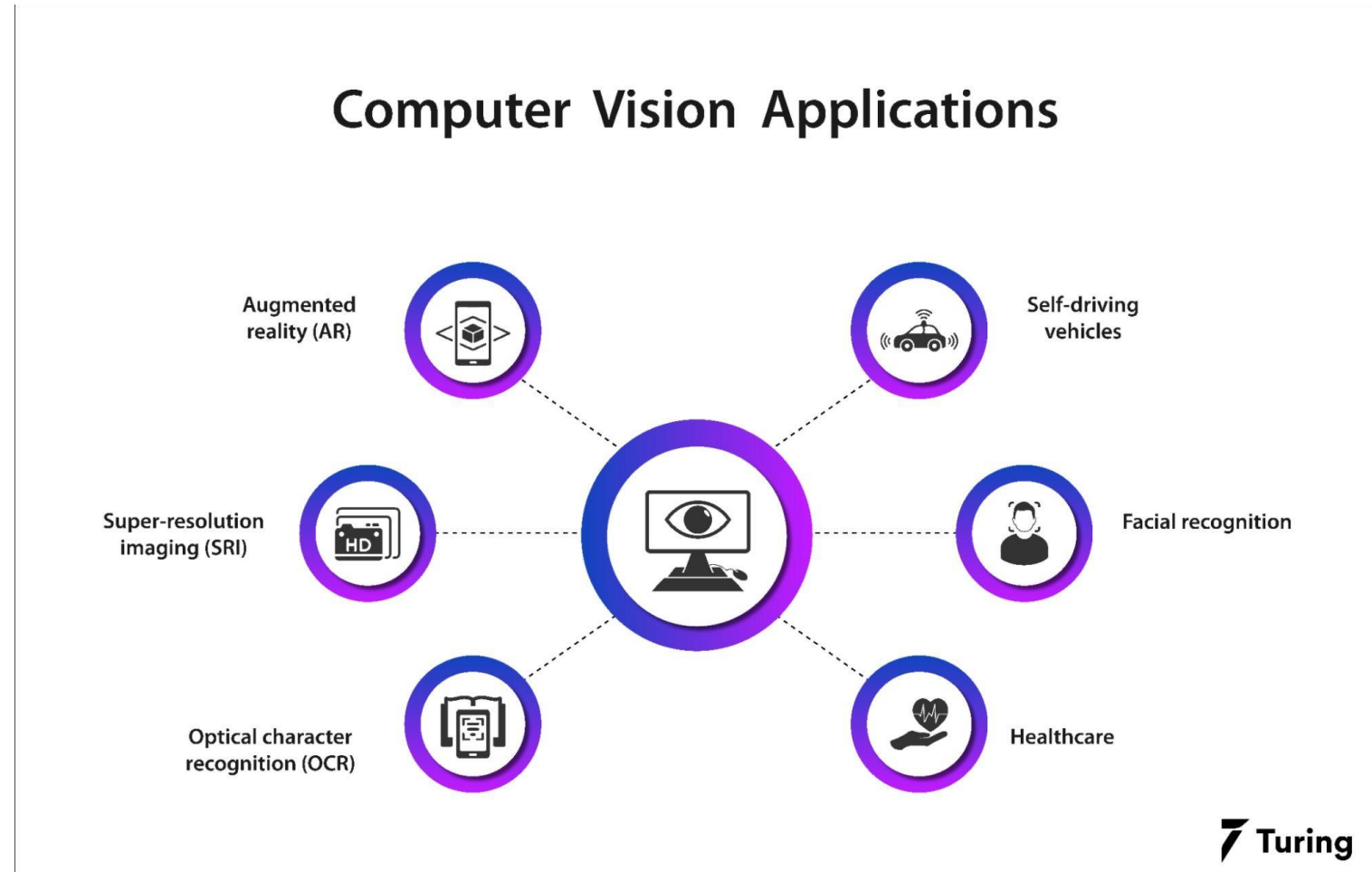
1.2. Concepts of computer vision?

Basic:

- Image Processing
- Pixel transformation
- Color Correction
- Padding
- Filtering

Advance:

- Object classification
- Object Detection
- Object Recognition
- Image Segmentation



2. Numpy And Image Basics

2.1. What is an image?

- How does a computer represent image data?
- Let's imagine we have a simple image of a handwritten number:

1 2 3

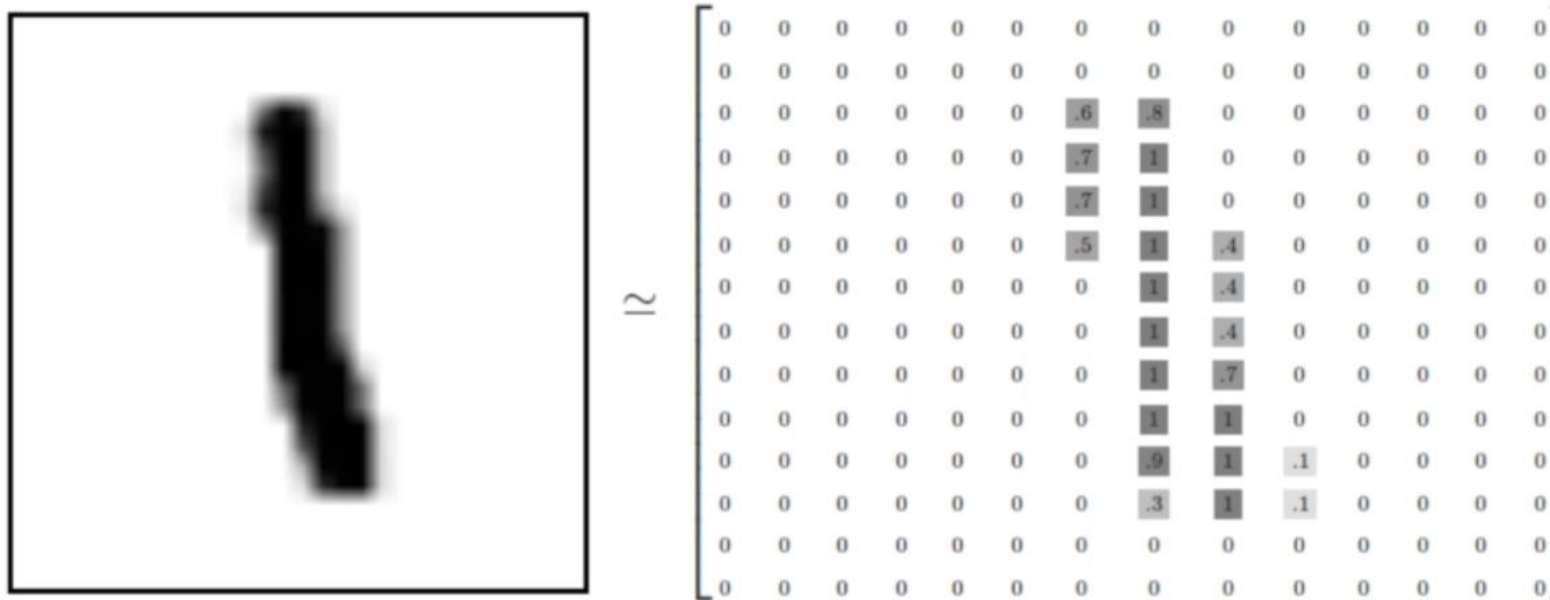
- Each single digit image can be represented as an array (grayscale images)



2. Numpy And Image Basics

2.1. What is an image?

- For example, here a number is 28 by 28 pixels

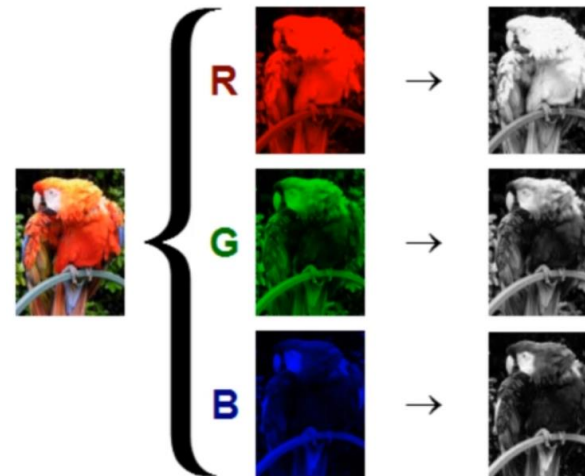
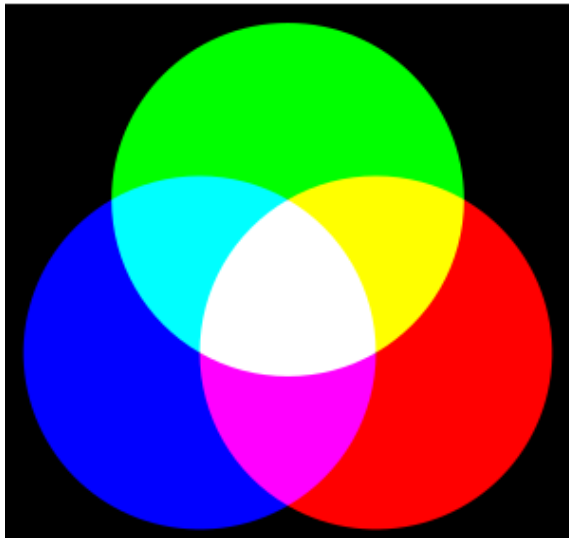


a **pixel** (abbreviated **px**), **pel**,^[1] or **picture element**^[2] is the smallest addressable element in a [raster image](#), or the smallest addressable element in a [dot matrix display device](#).

2. Numpy And Image Basics

2.1. What is an image?

- What about color images?
- Color images can be represented as a combination of Red, Green, and Blue

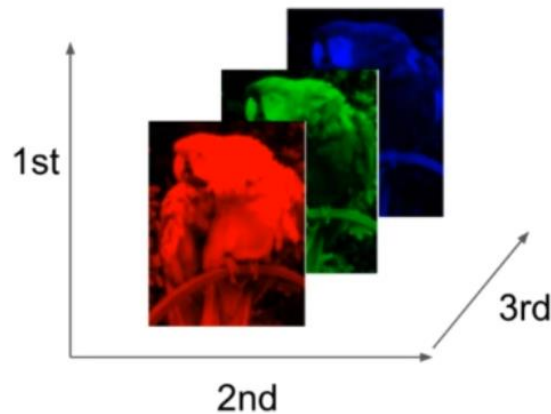


- The shape of the color array then has 3 dimensions.
- Height
- Width
- Color Channels

2. Numpy And Image Basics

2.1. What is an image?

- This means when you read in an image and check its shape, it will look something like:
- **(1280,720, 3)**
- **1280** pixel width
- **720** pixel height
- **3** color channels



$$\begin{bmatrix} 100 & 101 & 131 \\ 150 & 10 & 111 \\ 10 & 200 & 100 \end{bmatrix}, \begin{bmatrix} 100 & 112 & 20 \\ 210 & 120 & 120 \\ 260 & 20 & 20 \end{bmatrix}, \begin{bmatrix} 50 & 3 & 80 \\ 130 & 130 & 130 \\ 30 & 30 & 3 \end{bmatrix}$$

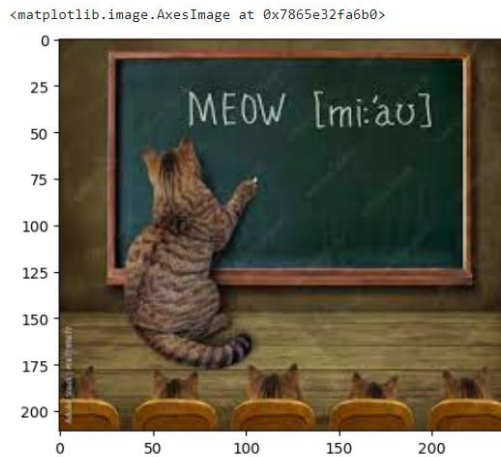
R **G** **B**

$$\begin{bmatrix} (r_{1,1}, g_{1,1}, b_{1,1}) & (r_{1,2}, g_{1,2}, b_{1,2}) & \dots & (r_{1,800}, g_{1,800}, b_{1,800}) \\ (r_{2,1}, g_{2,1}, b_{2,1}) & (r_{2,2}, g_{2,2}, b_{2,2}) & \dots & (r_{2,800}, g_{2,800}, b_{2,800}) \\ \dots & \dots & \dots & \dots \\ (r_{600,1}, g_{600,1}, b_{600,1}) & (r_{600,2}, g_{600,2}, b_{600,2}) & \dots & (r_{600,800}, g_{600,800}, b_{600,800}) \end{bmatrix}$$

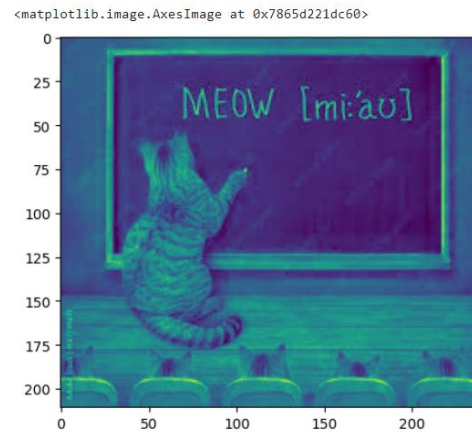
2. Numpy And Image Basics

2.2. Images and Numpy

```
9] plt.imshow(meow_arr)
```



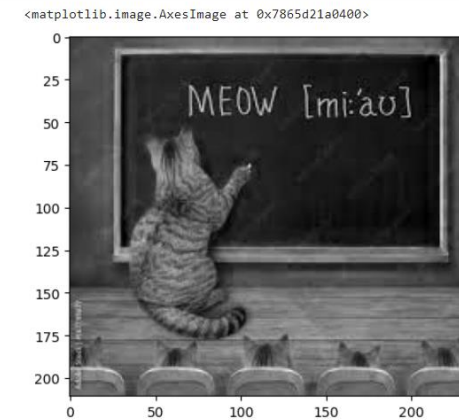
```
[16] plt.imshow(red_meo[:, :, 0])
```



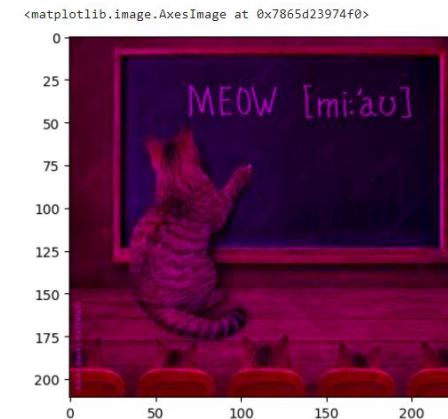
```
[17] red_meo[:, :, 0]
```

```
array([[ 65,  64,  64, ...,  58,  52,  45],  
       [ 65,  64,  64, ...,  58,  53,  48],  
       [ 64,  64,  65, ...,  58,  56,  53],  
       ...,  
       [106, 106, 105, ..., 106, 102,  99],  
       [106, 106, 107, ..., 106, 103,  99],  
       [106, 106, 107, ..., 106, 103,  99]], dtype=uint8)
```

```
[20] plt.imshow(red_meo[:, :, 0], cmap='gray')
```



```
[21] # green channel  
red_meo[:, :, 1] = 0  
plt.imshow(red_meo)
```



2. Numpy And Image Basics

2.2. Images and Numpy

Lab: https://drive.google.com/file/d/1qM4VneE6STM_j9KHy5EI1LVgV_30opco/view?usp=drive_link

3. Image Basics with OpenCV

3.1. Introduction

- OpenCV (Open Source Computer Vision) is a library of programming functions mainly aimed at real-time computer vision
- Created by Intel in 1999, It is written in C++ (we will be using the Python bindings)



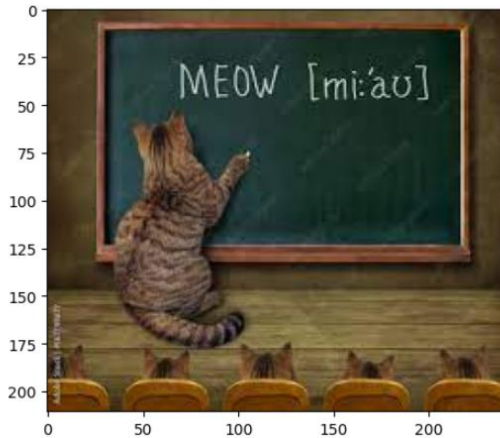
Section Goals:

- Be able to open image files with OpenCV
- Draw simple geometries on images
- Directly interact with an image through callbacks

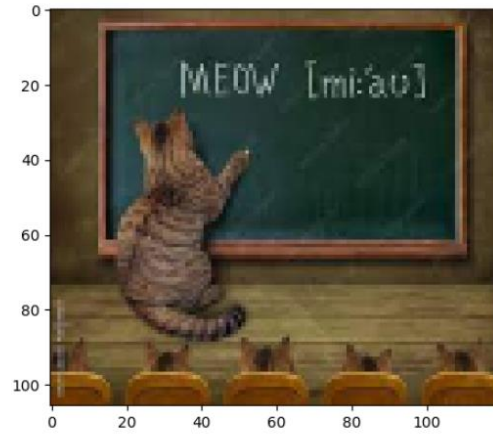
3. Image Basics with OpenCV

3.2. Open Image Files

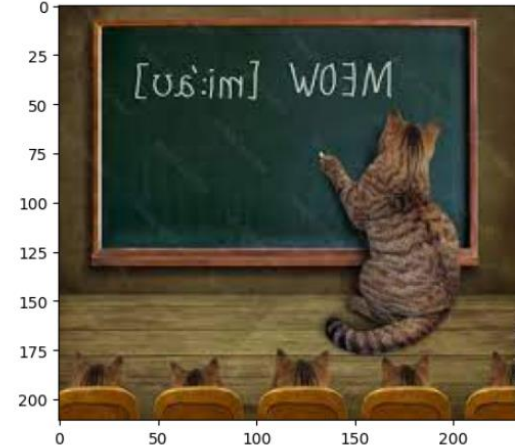
<matplotlib.image.AxesImage at 0x7a473b1a6980>



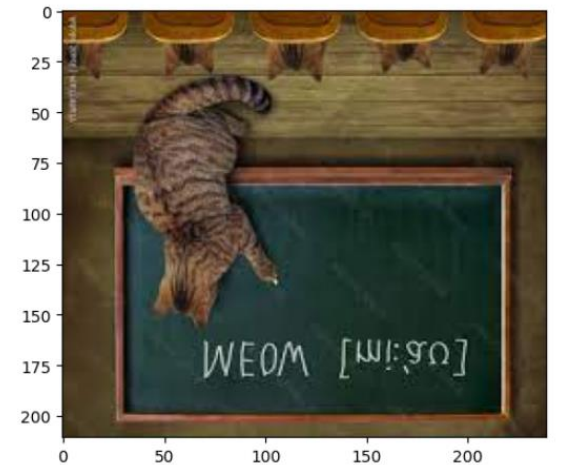
<matplotlib.image.AxesImage at 0x/a4/3a5/940>



<matplotlib.image.AxesImage at 0x7a4737a379a0>



<matplotlib.image.AxesImage at 0x7a4737a42f80>

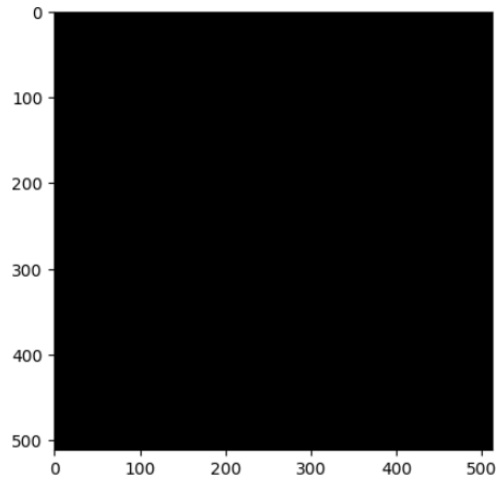


Lab: <https://drive.google.com/file/d/10A8gUu5bzxLSDzImquovQ9rQbsc4POSt/view?usp=sharing>

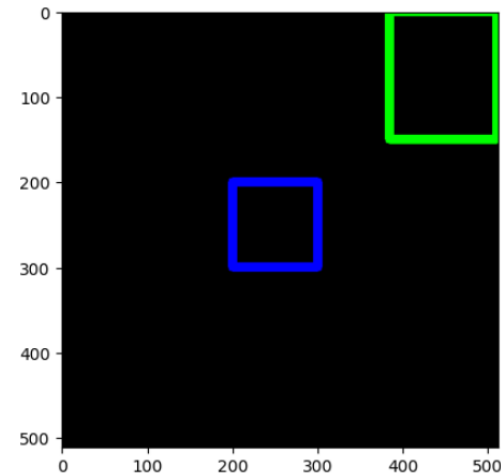
3. Image Basics with OpenCV

3.3. Drawing on Images

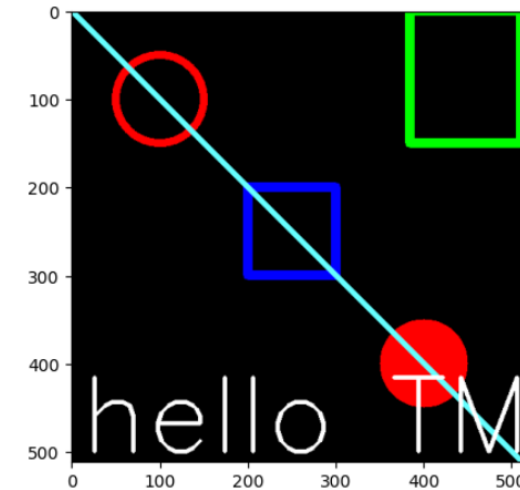
<matplotlib.image.AxesImage at 0x/b0db3c0tc/0>



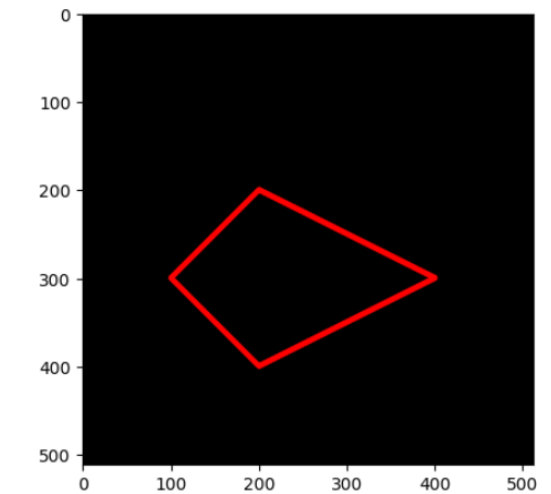
matplotlib.image.AxesImage at 0x7b0db334f3d0>



<matplotlib.image.AxesImage at 0x7ce82be43760>



<matplotlib.image.AxesImage at 0x7ce82bbb0220>

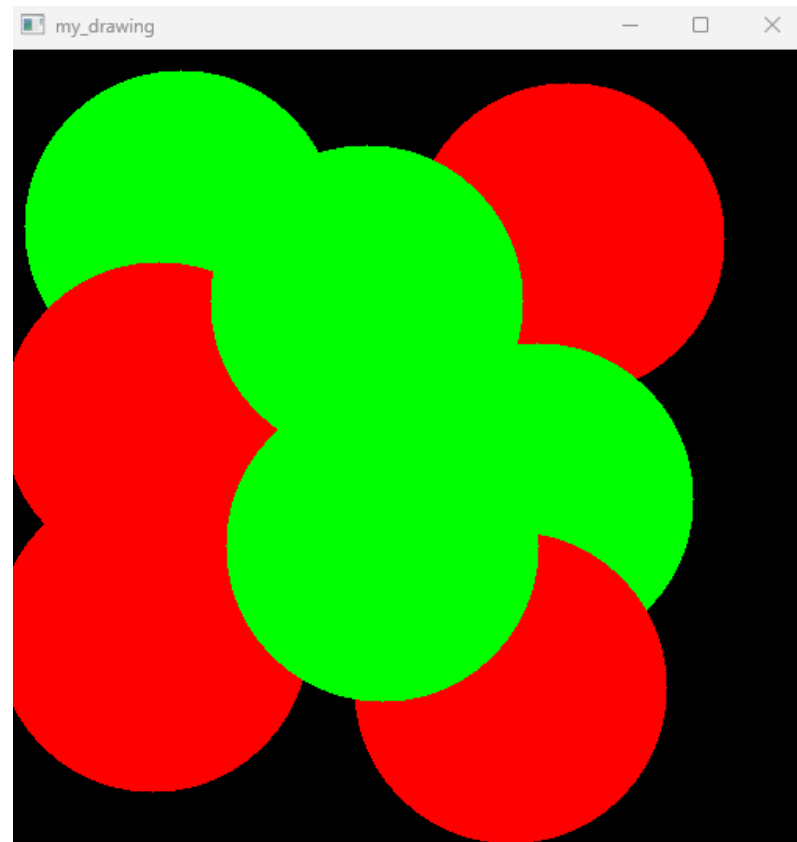


Lab: <https://drive.google.com/file/d/10A8gUu5bzxLSDzImquovQ9rQbsc4POSt/view?usp=sharing>

3. Image Basics with OpenCV

3.3. Direct Drawing on Images with a mouse

- Can use Callback to connect images to event functions with OpenCV
- This allow us to directly interact with images (and later on video)



3. Image Basics with OpenCV

Lab: Image basics assessment

<https://drive.google.com/file/d/10A8gUu5bzxLSDzImquovQ9rQbsc4POSt/view?usp=sharing>

4. Image Processing

4.1. Introduction

Section goals:

- Learn various image processing operations
- Perform image operations such as Smoothing, Blurring, Morphological Operations
- Grab properties such as color spaces and histograms

4. Image Processing

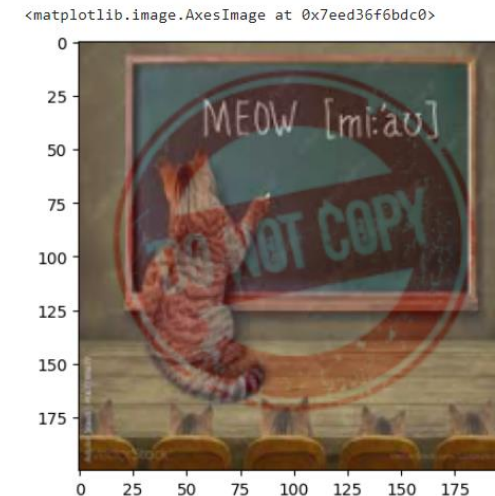
4.2. Blending Images

- Blending images is done through the **addWeighted** function that uses both images and combines them
- To blend images we use a simple formula:

○ $\text{New_pixel} = \alpha \times \text{pixel_1} + \beta \times \text{pixel_2} + \gamma$

```
[14] blended = cv2.addWeighted(src1=img1,alpha=0.8,src2=img2,beta=0.2,gamma=0)
```

```
plt.imshow(blended)
```



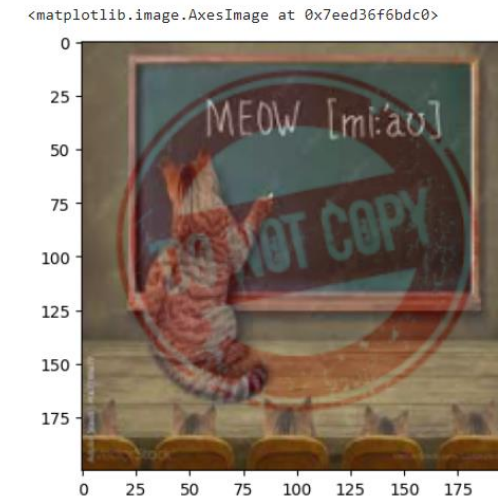
4. Image Processing

4.2. Images thresholding

- Blending images is done through the **addWeighted** function that uses both images and combines them
- To blend images we use a simple formula:
 - $\text{New_pixel} = \alpha \times \text{pixel_1} + \beta \times \text{pixel_2} + \gamma$

```
[14] blended = cv2.addWeighted(src1=img1,alpha=0.8,src2=img2,beta=0.2,gamma=0)
```

```
plt.imshow(blended)
```



4. Image Processing

Lab: https://drive.google.com/file/d/1tV3iRxVXhNtM_ejKVfjmGWG2uj_oUys0h/view?usp=drive_link

THANK YOU!



Email: contact@tmainnovation.vn



Website: www.tmainnovation.vn

